GT-ME-HND-M-21-002 YAKUBU

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MEC 414 = MEP 415 $ MEM 415 VOLUME (A)

ENGINEERING MATERIALS AND APPLICATIONS

USEFUL TERMINOLOGIES IN METALLURGY $ MATERIAL ENGINEERING

1. ENGINEERING = is the application of science by using your initiative and creativity to design, plan, construct and maintain building structure, machines and other manufactured goods
2. Metallurgy = is the study of chemical reaction that takes place in the production of metal and its alloys such as physical, chemical and mechanical behaviour of metal.
3. Metallurgical Engineering = is the branch of engineering that deals with the study of the structure and properties of metal, their extraction from the ground and the procedures for refining, alloying and making things from them.
4. Material Science = is the study of characteristics of nature, behaviour and uses of various materials such as plastics and metals as applied to science and technology
5. Material Engineering = is the branch of engineering that deals with the study of processing materials such as glass, plastics and materials to meet the demand of the users.
6. Mechanical Metallurgy: is the study of science and technology of metal behaviours in relation to mechanical force impose on them. This includes rocking, extrusion, bending and deep drawing
7. Production Metallurgy = is the planning and controlling of mechanical means of changing the shape, condition and relationship of material towards greater effectiveness and value
8. Extractive Metallurgy = is the branch of metallurgy that deals with the geological survey being carried out on mineral

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(9) Chemical Metallurgy = the branch of metallurgy that deal with reaction taking place within the context of mineral ore while various chemicals are being added. This aspect lea to the extraction of pure minerals from its ores.

ALLOY = is a substance that has mixture of two or more metals or mixture of a metal with a non-metallic material. Commerciall pure metals find many valuable uses such as Copper for conducti Aluminium for foil, and Chromium for plate surface. Simple carbo steels consist of about 0.5% Manganese, 0.8% Carbon and 98.7% Iron.

An alloy may consist of inter-metallic compound, a solid solution, an intimate mixture of minute crystals of the constituent metallic elements, or any combination of solutions or mixtures of different metals. Inter-metallic compounds such as Na-Au_2, Sn-Cu and Cu_3-Al_2 do not follow the ordinary rules of valence. They (inter-metallic compound) are generally hard and brittle.

Steel is stronger and harder than wrought iron, which is approximately pure iron and is used in far greater quantities. The alloy steels, mixtures of steel with chromium, manganese, molybdenum, nickel, tungsten and vanadium are stronger and harder than steel itself, and many of them are more corrosion-resistant than iron or steel. An alloy can often be made to match a pre-determined set of characteristics. An important case in which particular characteristics are necessary is the design of rockets, space-craft, and supersonic aircraft. The materials used in these vehicles and their engines must be light in weight, very strong, and able to sustain very high temperatures. To withstand these high temp[illegible]

Classification of Engineering materials
1- ferrous metals
2- non ferrous metal (Aluminum, magnisium, copper, nikel, titanium)
3- plastics (thermoplastics, thermoset)
4- Ceramics and diamond
5- Composite materials
6- Nano materials

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and titanium have been developed. To resist the heat generated du[ring]
operation into the atmosphere of the earth, alloys containing hea[t]
resistant metals such as tantalum, niobium, tungsten, cobalt, nick[el]
are being used in space vehicles (rocket and concorde airpla[ne]).
A wide variety of special alloys containing metals such as beryl[lium],
boron, niobium, hafnium and Zirconium, which have particular nuc[lear]
absorption characteristics, are used in nuclear reactors. Niobium-
alloys are used as superconductors at extremely low temperature.
Special copper, nickel, and titanium alloys, designed to resist th[e]
corrosive effects of boiling salt water, are used in desalination
plants.

Classification of Engineering Materials: can be broadly classified
as follows: (a) Ferrous Metals (b) Non-ferrous Metals (aluminum,
Magnesium, copper, nickel, titanium) (c) Plastics (thermoplastics, ther-
mosets) (d) Ceramics and Diamond (e) Composite Materials
(f) Nano Materials.

NOTE: the engineering materials are often primarily selected
based on their mechanical, physical, chemical and manufacturing
properties.
The secondary factors to be considered are the cost and availa-
bility, appearance, service life and recyclability.

Physical Properties of Metals: physical properties of elements are de-
pendent on the following factors: (a) the arrangement of their ato[ms]
or molecules in crystal lattice when in solids state (b) the bonds
that bind the atom or molecule in solid, liquid and gaseous state.
Metals are solid at room temperature and exist as crystal lattices, [in]
which their atoms are held together by strong metallic bonds me[...]

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PHYSICAL PROPERTIES: (1) High boiling and melting points (2) Conduc heat and electricity (3) Have characteristic luster (4) Relative high density

MECHANICAL PROPERTIES: (1) Ductile; that is easily drawn into wires (2) Possess high tensile strength (3) Malleable, that is easily hammered into sheets (4) Hard but not brittle (easily broken) with great tensile strength

However, some metals do not exhibits all these properties for example mercury is a liquid metal with a freezing point of $-39°C$ and boiling point of $357°C/360°C$ while sodium and potassium are light soft metals with low melting point of $97°C$ and $63°C$ respectively. Certain elements like Boron, Silicon and Germanium possess some properties falling between metals and non-metals, which is referred as METALLOIDS. Many metals are used in the "Pure" state for electroplating, durable goods, tools and electrical works or devices which include Cr, Ni, Cd, Sn, Zn, Os, Re and Rh.

Among the non-metals, carbon has an expectional characteristics as it is extremely hard with a very high melting point when it exists as diamond and it is a good conductor of electricity when it exists as graphite. Carbon is the brain of organic chemistry called hydrocarbon. Carbon is used in industrial application for motor brush and wear parts. Inert gases and other elements of non-metals are used in the elemental form for industrial application for prospective atmosphere.

Chemical Properties of Metals: the following are chemical properties of metal, which include:

(1) It acts as a reducing agent
(2) It reacts with acids to liberate hydrogen and salts
For instance $2HCl + 2Zn \longrightarrow ZnCl_2 + H_2 \uparrow$

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3 Metal ionizes very easily by losing its outermost electrons.
$Mg \rightarrow Mg^{2+} + 2e^-$ (Divalent) \$ $Al \rightarrow Al^{3+} + 3e^-$ (Trivalent)

VOLUME 2

by
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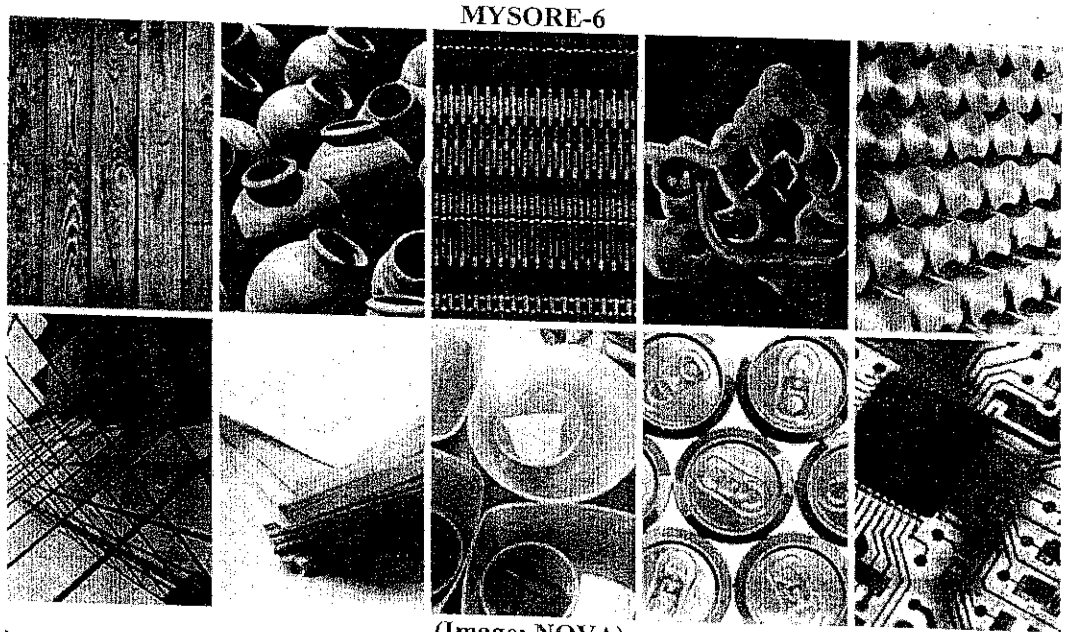

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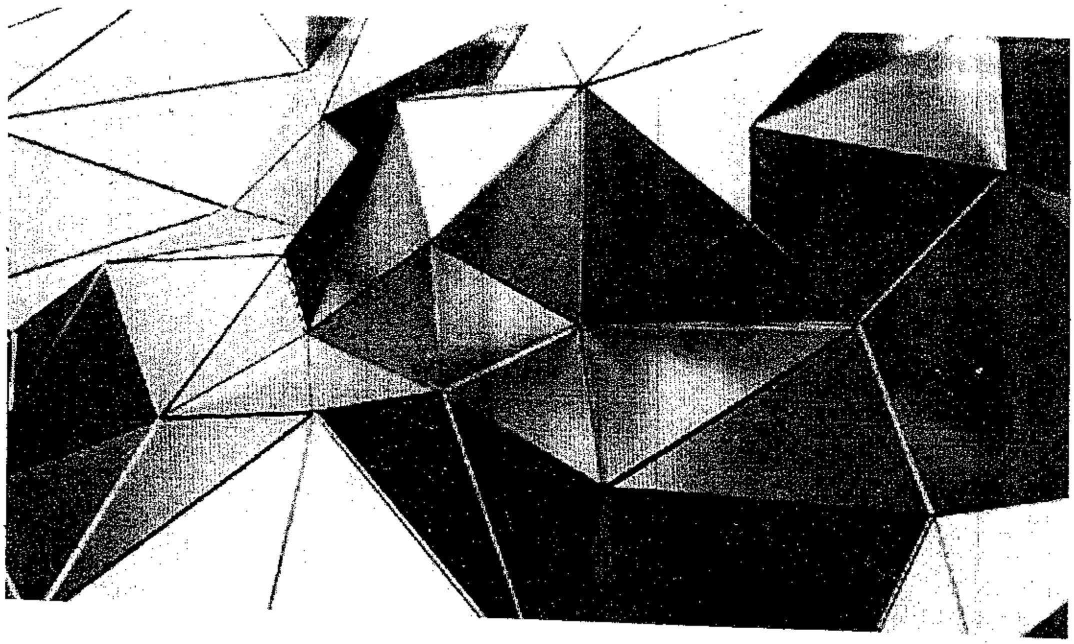

ET-ME-HND-M-21-002 YAKUBU

Introduction:

Earth's Materials have found to have played a dominant role in the development of civilization that we associate with them in all ages. In the origin of human life on Earth; the Stone Age, people used only natural materials, like stone, clay, skins, and wood. When people found copper and how to make it harder by alloying, the Bronze Age started about 3000 BC. The use of iron and steel, a stronger material that gave advantage in wars started at about 1200 BC. Understanding of how materials behave like they do, and why they differ in properties was only possible with the atomistic understanding allowed by quantum mechanics, that first explained atoms and then solids starting in the 1930s. The combination of physics, chemistry, and the focus on the relationship between the properties of a material and its microstructure is the domain of Materials Science. The development of this science allowed designing materials and provided a knowledge base for the engineering applications (Materials Engineering).

This report highlights the classification of the engineering materials and their processing techniques. The engineering materials can broadly be classified as:
a) Ferrous Metals
b) Non-ferrous Metals (aluminum, magnesium, copper, nickel, titanium)
c) Plastics (thermoplastics, thermosets)
d) Ceramics and Diamond
e) Composite Materials & f) Nano-materials.
The engineering materials are often primarily selected based on their mechanical, physical, chemical and manufacturing properties.

The secondary factors to be considered are the cost and availability, appearance, service life and recyclability.

Metals :

Metals are usually lustrous, ductile, malleable, and good conductors of electricity
They are divided into 2 categories:
A) FERROUS: the group which contains mainly iron (Fe). Iron is the most important metal in industrialized countries
B) NON-FERROUS: other metallic materials containing no iron likecopper (Cu) or aluminium (Al)

- Ferrous metals and alloys (irons, carbon steels, alloy steels, stainless steels, tool and die steels)
- Nonferrous metals and alloys (aluminum, copper, magnesium, nickel, titanium, precious metals, refractory metals, super-alloys)

All the elements are broadly divided into metals and non-metals according to their properties.

Metals are element substances which readily give up electrons to form metallic bonds and conduct electricity. Some of the important basic properties of metals are:
(a) metals are usually good electrical and thermal conductors,
(b) at ordinary temperature metals are usually solid,
(c) to some extent metals are malleable and ductile,
(d) the freshly cut surfaces of metals are lustrous,
(e) when struck metal produce typical sound, and
(f) most of the metals form alloys. When two or more pure metals are melted together to form a new metal whose properties are quite different from those of original metals, it is called an alloy.

Metallic materials possess specific properties like plasticity and strength.
Few favourable characteristics of metallic materials are high lustre, hardness, resistance to corrosion, good thermal and electrical conductivity, malleability, stiffness, the property of magnetism, etc.
Metals may be magnetic, non-magnetic in nature.
These properties of metallic materials are due to:
(i) the atoms of which these metallic materials are composed and
(ii) the way in which these atoms are arranged in the space lattice.

Metallic materials are typically classified according to their use in engineering as under:
Pure Metals and alloys.

Pure Metals:

Generally it is very difficult to obtain pure metal. Usually, they are obtained by refining the ore. Mostly, pure metals are not of any use to the engineers. Materials in this group are composed of one or more metallic elements (such as iron, aluminum, copper, titanium, gold, and nickel), and often also nonmetallic elements (for example, carbon, nitrogen, and oxygen) in relatively small amounts. Atoms in metals and their alloys are arranged in a very orderly manner and in comparison to the ceramics and polymers, are relatively dense. With regard to mechanical characteristics, these materials are relatively stiff, and strong yet are ductile (i.e., capable of large amounts of deformation without fracture), and are resistant to fracture , which accounts for their widespread use in structural applications.

Metallic materials have large numbers of nonlocalized electrons; that is, these electrons are not bound to particular atoms. Many properties of metals are directly attributable to these electrons. For example, metals are extremely good conductors of electricity and heat, and are not transparent to visible light; a polished metal surface has a lustrous appearance.
In addition, some of the metals (viz., Fe, Co, and Ni) have desirable magnetic properties.

Metals:

Normally metallic materials are combinations of metallic elements. Metallic materials have large number of nonlocalized electrons, i.e. electrons are not bound to particular atoms. Many properties of metals are directly attributable to these electrons. All metals are characterized by metallic properties, e.g. luster, opacity, malleability, ductility and electrical conductivity. Although metals compose about three fourth of the known elements but few find service in their pure form. The desired properties for engineering purposes are often found in alloys. Typical examples of metallic materials are iron, aluminium, copper, zinc, etc. and their alloys. They can be used either in bulk or powder form. Metals are extremely good conductors of electricity and heat are not transparent to visible light; a polished metal surface has a lustrous appearance. Moreover, metals are quite strong, yet deformable, which accounts for their extensive use in structural applications.

Metallic materials are always crystalline in nature. Scientists have developed amorphous (non-crystalline) alloys by very rapid cooling of a melt or by very high-energy mechanical milling. Recently, scientists have developed materials through rapid solidification called as quasicrystals. These are neither crystalline nor amorphous, but form an ordered structure somewhere between two known structures. These materials are expected to exhibit far reaching electrical properties.

Inorganic Materials:

These materials include metals, clays, sand rocks, gravels, minerals and ceramics and have mineral origin.
These materials are formed due to natural growth and development of living organisms and are not biological materials.
Rocks are the units which form the crust of the earth.
The three major groups of rocks are:

(i) Igneous Rocks: These rocks are formed by the consolidation of semi-liquid of liquid material (magma).

These are called as Plutonic if their consolidation takes place deep within the earth and volcanic if lava or magma solidifies on the earth's surface.

Basalt is igneous volcanic where as granite is igneous plutonic.

(ii) Sedimentary Rocks:
When broken down remains of existing rocks are consolidated under pressure, then the rocks are named as sedimentary rocks, e.g., shale and sandstone rocks.

The required pressure for the formation of sedimentary rocks is supplied by the overlying rocky material.

(iii) Metamorphic Rocks:
These rocks are basically sedimentary rocks which are changed into new rocks by intense heat and pressure, e.g., marble and slates. The structure of these rocks is in between igneous rocks and sedimentary rocks.

Rock materials are widely used for the construction of buildings, houses, bridges, monuments, arches, tombs, etc. The slate, which has got great hardness is still used as roofing material. Basalt, dolerite and rhyolite are crushed into stones and used as concrete aggregate and road construction material.

Another type of materials, i.e. Pozzolanics, are of particular interest to engineers because they are naturally occurring or synthetic silicious materials which hydrate to form cement. Volcanic ash, blast furnace slag, some shales and fly ash are examples of pozzolanic materials. When the cement contains 10- 20% ground blast furnace slag, then it is called pozzolans-portland cement, which sets more slowly than ordinary portland cement and has greater resistance to sulphate solutions and sea water.

Rocks, stone, wood, copper, silver, gold etc. are the naturally occurring materials exist in nature in the form in which they are to be used. However, naturally occurring materials are not many in number. Nowadays, most of the materials are manufactured as per requirements. Obviously, the study of engineering materials is also related with the manufacturing process by which the materials are produced to acquire the properties as per requirement.

Copper, silver, gold, etc. metals, which occur in nature, in their free state are mostly chemically inert and highly malleable and ductile as well as extremely corrosion resistant.

Alloys of these metals are harder than the basic metals. Carbonates, sulphates and sulphide ores are more reactive metals.

Biological Materials

Leather, limestone, bone, horn, wax, wood etc. are biological materials. Wood is fibrous composition of hydrocarbon, cellulose and lignin and is used for many purposes.

Apart from these components a small amount of gum, starch, resins, wax and organic acids are also present in wood. One can classify wood as soft wood and hard wood. Fresh wood contains high percentage of water and to dry out it, seasoning is done. If proper seasoning is not done, defects such as cracks, twist, wrap etc. may occur. Leather is obtained from the skin of animals after cleaning and tanning operations.

Nowadays, it is used for making belts, boxes, shoes, purses etc. To preserve the leather, tanning is used.

Following two tanning techniques are widely used:

(a) Vegetable Tanning: It consist of soaking the skin in tanning liquor for several days and then dried to optimum conditions of leather.

(b) Chrome Tanning: This technique involves pickling the skin in acid solution and then revolving in a drum which contains chromium salt solution. After that the leather is dried and rolled. Limestone is an important material which is not organic but has biological origin. It mainly consist of calcium carbonate and limestone. It is widely used to manufacture cement. In Iron and Steel Industries, limestone in pure form is used as flux. In early days bones of animals were used to make tools and weapons. Nowadays bones are used for the manufacture of glue, gelatin etc. Bones are laminate of organic substances and phosphates and carbonates of calcium. These are stronger in compression as compared to tension.

Organic Materials

Organic materials are carbon compounds and their derivatives. They are solids composed of long molecular chains. The study of organic compounds is very important because all biological systems are composed of carbon compounds. There are also some materials of biological origin which do not possess organic composition, e.g., limestone.
These materials are carbon compounds in which carbon is chemically bonded with hydrogen, oxygen and other non-metallic substances. The structure of these compounds is complex.
Common organic materials are
plastics and synthetic rubbers which are termed as organic polymers.

Other examples of organic materials
are wood, many types of waxes and petroleum derivatives.

Organic polymers are prepared by polymerisation reactions, in which simple molecules are chemically combined into long chain molecules or three-dimensional structures.
Organic polymers are solids composed of long molecular chains. These materials have low specific gravity and good strength.

The two important classes of organic polymers are:
(a) Thermoplastics: On heating, these materials become soft and hardened again upon cooling, e.g., nylon,polythene, etc.
(b) Thermosetting plastics: These materials cannot be resoftened after polymerisation, e.g., urea-formaldehyde,
phenol formaldehyde, etc. Due to cross-linking, these materials are hard, tough, non-swelling and brittle. These materials are ideal for moulding and casting into components. They have good corrosion resistance.

The excellent resistance to corrosion, ease of fabrication into desired shape and size, fine lusture, light weight, strength, rigidity have established the polymeric materials and these materials are fast replacing many metallic components.
PVC (Polyvinyl Chloride) and polycarbonate polymers are widely used for glazing, roofing and cladding of buildings.

Plastics are also used for reducing weight of mobile objects, e.g., cars, aircrafts and rockets. Polypropylenes and polyethylene are used in pipes and manufacturing of tanks.

Thermo-plastic films are widely used as lining to avoid seepage of water in canals and lagoons. To protect metal structure from corrosion, plastics are used as surface coatings.

Plastics are also used as main ingredients of adhesives. The lower hardness of plastic materials compared with other materials makes them subjective to attack by insects and rodents.

Because of the presence of carbon, plastics are combustible. The maximum service temperature is of the order of 100°C. These materials are used as thermal insulators because of lower thermal conductivity. Plastic materials have low modulus of rigidity, which can be improved by addition of filters, e.g., glass fibres. Natural rubber, which is an organic material of biological origin, is an thermoplastic material. It is prepared from a fluid, provided by the rubber trees. Rubber materials are widely used for tyres of automobiles, insulation of metal components, toys and other rubber products.

Polymers
Polymers include the familiar plastic and rubber materials. Many of them are organic compounds that are chemically based on carbon, hydrogen, and other nonmetallic elements (viz.O,N, and Si).

Furthermore, they have very large molecular structures, often chain-like in nature that have a backbone of carbon atoms.
Some of the common and familiar polymers are polyethylene (PE),
nylon,
poly(vinyl chloride) (PVC),
polycarbonate (PC), polystyrene (PS), and silicone rubber.
These materials typically have low densities , whereas their mechanical characteristics are generally dissimilar to the metallic and ceramic materials—they are not as stiff nor as strong as these other material types.

However, on the basis of their low densities,many times their stiffnesses and strengths on a per mass basis are comparable to the metals and ceramics.

In addition, many of the polymers are extremely ductile and pliable (i.e., plastic), which means they are easily formed into complex shapes. In general, they are relatively inert chemically and unreactive in a large number of environments.

One major drawback to the polymers is their tendency to soften and/or decompose at modest temperatures, which, in some instances, limits their use.
Furthermore, they have low electrical conductivities and are nonmagnetic. Fiberglass is sometimes also termed a "glass fiber-reinforced polymer" composite, abbreviated "GFRP."
Many of these are organic substances and derivatives of carbon and hydrogen. Polymers include the familiar plastic and rubber materials.
Usually polymers are classified into three categories: thermoplastic polymers, thermosetting polymers and elastomers, better called as rubbers.
Polymers have very large molecular structures. Most plastic polymers are light in weight and are soft in comparison to metals.
Polymer materials have typically low densities and may be extremely flexible and widely used as insulators, both thermal and electrical.
Typical examples of polymers are polyesters, phenolics, polyethylene, nylon and rubber.

The overriding consideration of the selection of a given polymer is whether or not the material can be processed into the required article easily and economically.

- Crude oil supplies the majority of the raw material for the production of polymers, also called plastics
- Polymers can be divided into 3 categories:

Thermoplastics: usually soft and easy to be recycled
Thermosetting plastics: usually stiff and not easy to be recycled.
Elastomers: flexible (rubbers)

Ceramic materials
The word ceramic is derived from the Greek word keramikos.
The term covers inorganic non -metallic materials whose formation is due to the action of heat Clays, bricks, cements, glass are the most important ones. These are crystalline compounds between metallic and non-metallic elements. They are most frequently oxides, nitrides and carbides.

Ceramics are compounds between metallic and nonmetallic elements; they are most frequently oxides, nitrides, and carbides.
For example, some of the common ceramic materials include
aluminum oxide (or alumina,Al_2O_3),
silicon dioxide (or silica, SiO_2),
silicon carbide (SiC)& silicon nitride (Si_3N_4).

In addition, some are referred to as the traditional ceramics—those composed of clay minerals (i.e., porcelain), as well as cement, and glass.

With regard to mechanical behavior, ceramic materials are relatively stiff and strong—stiffnesses and strengths are comparable to those of the metals. In addition, ceramics are typically very hard.
On the other hand, they are extremely brittle (lack ductility), and are highly susceptible to fracture.

These materials are typically insulative to the passage of heat and electricity (i.e., have low electrical conductivities, and are more resistant to high temperatures and harsh environments than metals and polymers.

With regard to optical characteristics, ceramics may be transparent, translucent, or opaque, and some of the oxide ceramics (e.g., Fe O) exhibit magnetic behavior.

Nowadays graphite is also categorized in ceramics. The wide range of materials which falls within this classification include ceramics that are composed of clay minerals, cement and glass.
Glass is grouped with this class because it has similar properties but most glasses are amorphous.
Ceramics are characterised by high hardness, abrasion resistance, brittleness and chemical inertness.
Ceramics are typically insulative to the passage of electricity and heat, and are more resistant to high temperatures and harsh environments than metals and polymers.
With regard to mechanical behaviour, these materials are hard but very brittle. These materials are widely categorized into oxide and non-oxide ceramics.
Examples **Ceramics:** Glasses; Glass ceramics; Graphite; Diamond.

Composites

A composite is a composition of two or more materials in the first three categories, e.g. metals, ceramics and polymers, that has properties from its constituents. Large number of composite materials have been engineered. Few typical examples of composite materials are wood, clad metals, fibre glass, reinforced plastics, cemented carbides, etc.
Fibre glass is a most familiar composite material, in which glass fibres are embedded within a polymeric material.

A composite is designed to display a combination of the best characteristics of each of the component materials. Fibre glass acquires strength from the glass and the flexibility from the polymer.
A composite is composed of two (or more) individual materials, which come from the categories discussed above—viz., metals, ceramics, and polymers. The design goal of a composite is to achieve a combination of properties that is not displayed by any single material, and also to incorporate the best characteristics of each of the component materials.

A large number of composite types exist that are represented by different combinations of metals, ceramics, and polymers. Furthermore, some naturally-occurring materials are also considered to be composites—for example, wood and bone. However, most of those we consider in our discussions are synthetic (or man-made) composites.

One of the most common and familiar composites is fiber-glass, in which small glass fibers are embedded within a polymeric material (normally an epoxy or polyester).

The glass fibers are relatively strong and stiff (but also brittle), whereas the polymer is ductile (but also weak and flexible). Thus, the resulting fiberglass is relatively stiff, strong, flexible, and ductile.

In addition, it has a low density. Another of these technologically important materials is the "carbon fiber-reinforced polymer" (or "CFRP") composite—carbon fibers that are embedded within a polymer.

These materials are stiffer and stronger than the glass fiber-reinforced materials, yet they are more expensive.
The CFRP composites are used in some aircraft and aerospace applications, as well as high-tech sporting equipment (e.g., bicycles, golf clubs, tennis rackets, and skis/snowboards).

A true composite structure should show matrix material completely surrounding its reinforcing material in which the two phases act together to exhibit desired characteristics. These materials as a class of engineering material provide almost an unlimited potential for higher strength, stiffness, and corrosion resistance over the 'pure' material systems of metals, ceramics and polymers.

Many of the recent developments of materials have involved composite materials. Probably, the composites will be the steels of this century.

Nowadays, the rapidly expanding field of nano composites is generating many exciting new materials with novel properties. The general class of nano composite organic or inorganic material is a fast growing field of research.
Significant efforts are going on to obtain control of nano composite materials depend not only on the properties of their individual parents but also on their morphology and interfacial characteristics.
The lamellar class of intercalated organic/ inorganic nano composites and namely those systems that exhibit electronic properties in at least one of the composites offers the possibility of obtaining well ordered systems some of which may lead to unusual electrical and mechanical properties.

Polymer-based nano composites are also being developed for electronic applications such as thin-film capacitors in integrated circuits and solid polymer electrolytes for batteries.
No doubt, the field of nano composites is of broad scientific interest with extremely impressive technological promise.

Other materials coming under this group are: Reinforced plastics;
Metal-matrix composites; Ceramic-matrix composites; Sandwich structures; and Concrete.

Advanced Materials

Materials that are utilized in high-technology (or high-tech) applications are sometimes termed advanced materials. By high technology we mean a device or product that operates or functions using relatively intricate and sophisticated principles; examples include electronic equipment (camcorders, CD/DVD players, etc.), computers, fiber-optic systems, spacecraft, aircraft, and military rocketry.

These advanced materials are typically traditional materials whose properties have been enhanced, and, also newly developed, high-performance materials.

Furthermore, they may be of all material types (e.g., metals, ceramics, polymers), and are normally expensive.

Advanced materials include semiconductors, biomaterials, and what we may term "materials of the future" (that is, smart materials and nanoengineered materials), which we discuss below. The properties and applications of a number of these advanced materials -for example, materials that are used for lasers, integrated circuits, magnetic information storage, liquid crystal displays (LCDs), and fiber optics.

These are new engineering materials which exhibit high strength, great hardness, and superior thermal, electrical, optical and chemical properties.
Advanced materials have dramatically altered communication technologies, reshaped data analysis, restructured medical devices, advanced space travel and transformed industrial production process.

These materials are often synthesized from the biproducts of conventional commodity materials and often possess following characteristics:

- These materials are created for specific purposes,
- These materials are highly processed and possess a high value-to weight ratio,
- These materials are developed and replaced with high frequency, and
- These materials are frequently combined into new composites.

Nowadays, there is considerable interest in making advanced materials that are usually graded by chemical composition, density or coefficient of thermal expansion of material or based on microstructural features, e.g. a particular arrangement of second-phase particles or fibres in a matrix.

Such materials are referred as functionally graded materials.

Instead of having a step function, one may strive to achieve a gradual change.
Such gradual change will reduce the chances of mechanical and thermal stresses, generally present otherwise. We may note that the concept of a functionally graded material is applicable to any material metal, polymer or ceramic.

Semiconductors:

These materials have electrical properties that are intermediate between electrical conductors and insulators..Moreover, the electrical characteristics of semiconducting materials are extremely sensitive to the presence of minute concentrations of impurity atoms; these concentrations may be controlled over very small spatial regions. Silicon, Germanium and some more compounds form the vast majority of semiconducting crystals.

Semiconductors have electrical properties that are intermediate between the electrical conductors (viz. metals and metal alloys) and insulators (viz. ceramics and polymers).

Furthermore, the electrical characteristics of these materials are extremely sensitive to the presence of minute concentrations of impurity atoms, for which the concentrations may be controlled over very small spatial regions.

Semiconductors have made possible the advent of integrated circuitry that has totally revolutionized the electronics and computer industries (not to mention our lives) over the past three decades. These semiconducting materials are used in a number of solid state devices, e.g. diodes, transistors, photoelectric devices, solar batteries, radiation detectors, thermistors and lasers. The semiconductors have made possible the advent of integrated circuitary that has completely revolutionized the electronics and computer industries.

Biomaterials:

Biomaterials are employed in components implanted into the human body for replacement of diseased or damaged body parts.These materials must not produce toxic substances and must be compatible with body tissues (i.e., must not cause adverse biological reactions).
All of the above materials—metals, ceramics, polymers, composites, and semiconductors—may be used as biomaterials.
Some biomaterials that are utilized in artificial hip replacements.

Materials of the Future- Smart Materials:

Smart (or intelligent) materials are a group of new and state-of-the-art materials now being developed that will have a significant influence on many of our technologies.
The adjective "smart" implies that these materials are able to sense changes in their environments and then respond to these changes in predetermined manners traits that are also found in living organisms.
In addition, this "smart" concept is being extended to rather sophisticated systems that consist of both smart and traditional materials. function).

Actuators may be called upon to change shape, position, natural frequency, or mechanical characteristics in response to changes in temperature, electric fields, and/or magnetic fields.

Four types of materials are commonly used for actuators:
shape memory alloys, piezoelectric ceramics, magnetostrictive materials, and electrorheological/ magnetorheological fluids.

Shape memory alloys are metals that, after having been deformed, revert back to their original shapes when temperature is changed.

Piezoelectric ceramics expand and contract in response to an applied electric field (or voltage); conversely, they also generate an electric field when their dimensions are altered.

The behavior of magnetostrictive materials is analogous to that of the piezoelectrics, except that they are responsive to magnetic fields. Also, electrorheological and magnetorheological fluids are liquids that experience dramatic changes in viscosity upon the application of electric and magnetic fields, respectively.

Materials/devices employed as sensors include optical fibers, piezoelectric materials (including some polymers), and microelectromechanical devices.

For example, one type of smart system is used in helicopters to reduce aerodynamic cockpit noise that is created by the rotating rotor blades.

Piezoelectric sensors inserted into the blades monitor blade stresses and deformations; feedback signals from these sensors are fed into a computer-controlled adaptive device, which generates noise-canceling antinoise.
Smart or intelligent materials form a group of new and state of art materials now being developed that will have a significant influence on many of present-day technologies.

The adjective 'smart' implies that these materials are able to sense changes in their environments and then respond to these changes in predetermined manners—traits that are also found in living organisms. In addition, the concept of smart materials is being extended to rather sophisticated systems that consist of both smart and traditional materials.

The field of smart materials attempts to combine the sensor (that detects an input signal), actuator (that performs a responsive and adaptive function) and the control circuit or as one integrated unit.
Acutators may be called upon to change shape, position, natural frequency, or mechanical characteristics in response to changes in temperature, electric fields, and or magnetic fields.

Usually, four types of materials are commonly used for actuators: shape memory alloys, piezoelectric ceramics, magnetostrictive materials, and electrorheological / magnetorheological fluids. Shape memory alloys are metals that, after having been deformed, revert back to their original shapes when temperature is changed.

Piezoelectric ceramics expand and contract in response to an applied electric field (or voltage); conversely these materials also generate an electric field when their dimensions are altered. The behaviour of magnetostrictive materials is analogous to that of the piezoelectric ceramic materials, except that they are responsive to magnetic fields.

Also, electrorheological and magnetorheological fluids are liquids that experience dramatic changes in viscocity upon application of electric and magnetic fields, respectively. The combined system of sensor, actuator and control circuit or as one IC unit, emulates a biological system. These are known as smart sensors, Micro System Technology (MST) or Micro Electro Mechanical Systems (MEMS). Materials / devices employed as sensors include optical fibres, piezoelectric materials (including some polymers) and MEMS.

For example, one type of smart system is used in helicopters to reduce aero-dynamic cockpit noise that is created by the rotating rotor blades.

Piezoelectric sensors inserted into the blades, monitor blade stresses and deformations; feedback signals from these sensors are fed into a computer controlled adaptive device, which generates noise cancelling antidose.

MEMS devices are small in size, light weight, low cost, reliable with large batch fabrication technology. They generally consist of sensors that gather environmental information such as pressure, temperature, acceleration, etc., integrated electronics to process the data collected and actuators to influence and control the environment in the desired manner.

The MEMS technology involves a large number of materials.
Silicon forms the backbone of these systems also due to its excellent mechanical properties as well as mature micro-fabrication technology including lithography, etching, and bonding.

Other materials having piezoelectric, piezoresistive, ferroelectric and other properties are widely used for sensing and actuating functions in conjunction with silicon.

Nano Materials

Nano-structured (NS) materials are defined as solids having microstructural features in the range of 1–100 nm (= (1–100) × 10–9 m) in at least in one dimension. These materials have outstanding mechanical and physical properties due to their extremely fine grain size and high grain boundary volume fraction.

Usually, the clusters of atoms consisting of typically hundreds to thousands on the nanometer scale are called as nanoclusters. These small group of atoms, in general, go by different names such as nano particles, nanocrystals, quantum dots and quantum boxes. Significant work in being carried out in the domain of nano-structured materials and nano tubes since they were found to have potential for high technology engineering applications.

Nano-structured materials exhibit properties which are quite different from their bulk properties. These materials contain a controlled morphology with atleast one nano scale dimension. Nano crystals, nano wires and nano tubes of a large number of inorganic materials have been synthesized and characterized in the last few years.

Some of the nano materials exhibit properties of potential technological value. This is particularly true for nano-structures of semiconducting materials such as metal chalcogenides and nitrides. The mixing of nano-particles with polymers to form composite materials has been practiced for decades.

For example, the clay reinforced resin known as Bakelite is the first mass-produced polymer-nanoparticle composites and fundamentally transformed the nature of practical household materials. Even before bakelite, nano composites were finding applications in the form of nano particle-toughened automobile tires prepared by blending carbon black, zinc oxide, and/or magnesium sulfate particles with vulcanized rubber.

Despite these early successes, the broad scientific community was not galvanized by nano composites until the early 1990s, when reports revealed that adding mica to nylon produced a five-fold increase in the yield and tensile strength of the material. Subsequent developments have further contributed to the surging interest in polymer–nano particle composites.

Nanoengineered Materials

Until very recent times the general procedure utilized by scientists to understand the chemistry and physics of materials has been to begin by studying large and complex structures, and then to investigate the fundamental building blocks of these structures that are smaller and simpler. This approach is sometimes termed "top-down" science.

However, with the advent of scanning probe microscopes, which permit observation of individual atoms and molecules, it has become possible to manipulate and move atoms and molecules to form new structures and, thus, design new materials that are built from simple atomic-level constituents (i.e., "materials by design").

This ability to carefully arrange atoms provides opportunities to develop mechanical, electrical, magnetic, and other properties that are not otherwise possible.

We call this the "bottom-up" approach, and the study of the properties of these materials is termed "nanotechnology"; the "nano" prefix denotes that the dimensions of these structural entities are on the order of a nanometer (10^-9 m)—as a rule, less than 100 nanometers (equivalent to approximately 500 atom diameters).

Materials produced out of nanoparticles have some special features, e.g.
(i) very high ductility (ii) very high hardness ~4 to 5 times more than usual conventional materials
(iii) transparent ceramics achievable
(iv) manipulation of colour
(v) extremely high coercivity magnets
(vi) developing conducting inks and polymers.

Material science has expanded from the traditional metallurgy and ceramics into new areas such as
electronic polymers,
complex fluids,
intelligent materials,
organic composites,
structural composites,
biomedical materials (for implants and other applications),
biomimetics,
artificial tissues,
biocompatible materials,
"auxetic" materials (which grow fatter when stretched),
elastomers,
dielectric ceramics (which yield thinner dielectric layers for more compact electronics),
ferroelectric films (for non-volatile memories), more efficient photovoltaic converters,
ceramic superconductors,
improved battery technologies,
self-assembling materials,
fuel cell materials,
optoelectronics,
artificial diamonds,
improved sensors (based on metal oxides, or conducting polymers),
grated light values, ceramic coatings in air (by plasma deposition),
electrostrictive polymers,
chemical—mechanical polishing,
alkali-metal thermoelectric converters, luminescent silicon,
planar optical displays without phosphors, MEMS, and super molecular materials.

However, there still remain technological challenges,
including the development of even more sophisticated and specialized materials, as well as consideration of the environmental impact of materials production.

Material scientists are interested in green approaches, by entering the field of environmental—biological science, by developing environmentally friendly processing techniques and by inventing more recyclable materials.

The following table shows the properties of materials to be considered for different applications:

Manufacturing processes	Functional requirements	Cost considerations	Operating parameters
Plasticity	Strength	Raw material	Pressure
Malleability	Hardness	Processing	Temperature
Ductility	Rigidity	Storage	Flow
Machinability	Toughness	Manpower	Type of material
Casting properties	Thermal conductivity	Special treatment	Corrosion requirements
Weldability	Fatigue	Inspection	Environment
Heat	Electrical treatment	Packaging properties	Protection from fire
Tooling	Creep	Inventory	Weathering
Surface finish	Aesthetic look	Taxes and custom duty	Biological effects

Materials science:

The combination of physics, chemistry, and the focus on the relationship between the properties of a material and its microstructure is the domain of Materials Science. The development of this science allowed designing materials and provided a knowledge base for the engineering applications (Materials Engineering).

Conclusion:

Materials play a central role in society. Beyond the physical and chemical properties of materials, their cultural properties have often been overlooked in anthropological studies: finished products have been perceived as 'social' yet the materials which comprise them are considered 'raw' or natural'. Human societies have always worked with materials. However, the customs and traditions surrounding this differ according to the place, the time and the material itself. Materials are rapidly emerging as one of the most important and exciting research fields globally and across disciplines.

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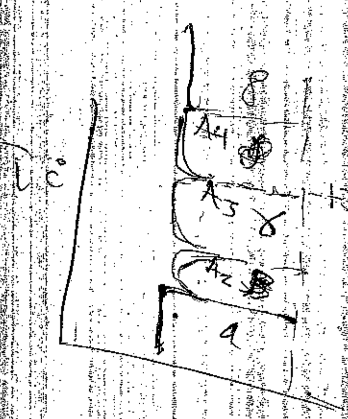
MEC 444 (= MEP 415 + MEM 415) VOLUME C 1

7 The Structure and Properties of Plain Carbon Steels and Cast Irons

(1)

7.1 The Allotropy of Iron

Iron and carbon exhibit complete liquid solubility but only partial solid solubility. The iron/carbon system includes several solid solutions and an intermediate phase, whilst the phase diagram shows peritectic, eutectic and eutectoid reactions. Before considering the structures of iron/carbon alloys it is essential to understand the structural changes that occur during the heating and cooling of pure iron itself.



Pure iron is **allotropic**, i.e. it exists in more than one physical form. At room temperature pure iron has a body-centred cubic crystal form and this is stable up to 910°C when it changes to a face-centred cubic structure. A further change takes place at 1390°C when the f.c.c. iron reverts to a b.c.c. form. Greek letters are used to denote these allotropic forms of iron as follows:

b.c.c. iron stable below 910°C	α Fe
f.c.c. iron stable between 910°C and 1390°C	γ Fe
b.c.c. iron stable above 1390°C	δ Fe

These structural changes in the solid state involve the evolution of heat during the cooling of iron from its freezing point (1535°C). These change points (called **critical points**) are readily apparent as arrests (A points) on an inverse rate cooling curve (fig. 7.1).

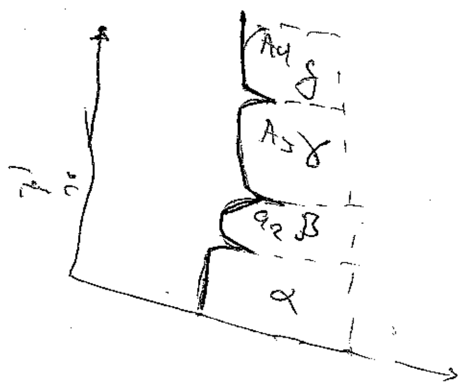
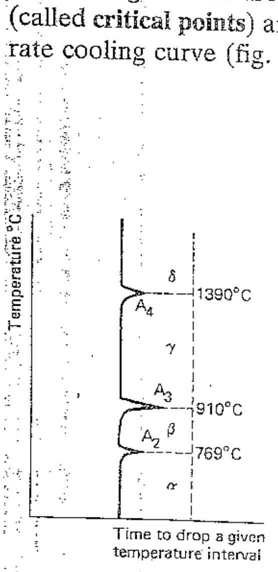


Fig. 7.1 Inverse rate cooling curve for pure iron



pure iron is allotropic, it exist in more than one form. At room temperature pure iron has a body centered cubic crystal form and this is stable up to 910°c when it changes to a face centred cubic

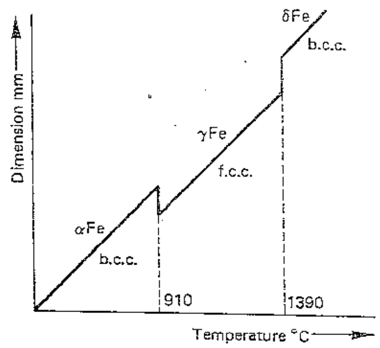


Fig. 7.2 Change in dimensions of pure iron with temperature

The atoms in the f.c.c. lattice are more densely packed together than in the b.c.c. form, so that in the change (on heating) from α iron to γ iron there is a contraction in volume; similarly when the reversion to the b.c.c. form takes place at the γ to δ critical point there is an expansion (fig. 7.2).

The critical points are referred to as follows:

the $\alpha \rightarrow \gamma$ change temperature is termed the A_3 point

the $\gamma \rightarrow \delta$ change temperature is termed the A_4 point.

Experimental determination shows that the temperatures at these change points are significantly different when heating cycles are used than when cooling cycles are employed. Thus, it is usual to refer to the change points obtained on heating as A_c points ("c" from chauffage), and those on cooling as A_r points ("r" from refroidissement). For example, the A_{c_3} and A_{r_3} points may be 935°C and 910°C respectively.

Another change which takes place in iron during heating is the loss of magnetism at 769°C (known as the Curie point). This temperature is referred to as the A_2 point, but no crystal change is involved and it is of no importance in heat-treatment.

The dimensional changes which take place during the alterations in crystal structure may be used as a basis for determining the temperatures at which the structural changes occur (i.e. the **critical temperatures**). The length of a small specimen of the iron is accurately measured over a known range of temperature using an instrument called a *dilatometer*. A graph of length as a function of temperature is plotted and the critical points are clearly apparent (fig. 7.2). Very slow heating and cooling rates must be used in order to minimise errors due to thermal lag.

7.2 The Iron/Carbon Phase Diagram

The addition of carbon to iron lowers the freezing point of the metal and also alters the temperatures at which the changes in crystal structure occur. Body-centred cubic iron has only very slight solid solubility for carbon to form a solid solution **ferrite** at low temperatures and δ-ferrite at high temperatures. Face-centred cubic iron can dissolve up to 1.7% carbon to form a solid

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soft, ductile and readily cold-worked. ... is also soft and ductile. Carbon also reacts with iron to form a hard, brittle compound, Fe_3C, which is called **cementite**.

The relevant phase diagram is the iron/cementite diagram, extending to 6.67% carbon (i.e. the carbon content of Fe_3C). The iron/cementite system is not in true equilibrium since the cementite can decompose, giving elemental carbon (graphite). The system is said to be in a meta-stable state. The phase diagram (fig. 7.3) may be considered as being made up of several portions, in which eutectic, eutectoid and peritectic reactions occur.

During cooling, the *peritectic* reaction occurs at 1490°C by δ solid solution, containing 0.07% carbon, reacting with liquid containing 0.55% carbon to form a new solid solution, austenite, containing 0.18% carbon. The peritectic reaction has no effect on the micro-structure which exists at room temperature and thus does not influence the properties.

The *eutectic* reaction occurs at 1130°C when liquid containing 4.3% carbon solidifies to form a mixture of austenite and cementite. This eutectic mixture is sometimes referred to as ledeburite.

The *eutectoid* reaction occurs at 723°C when austenite containing 0.8% carbon decomposes to give a structure consisting of alternate layers of ferrite and cementite. This eutectoid structure is known as **pearlite** due to its "mother of pearl" appearance.

Taking the relative atomic masses of iron and carbon as 56 and 12 respectively, the proportions of ferrite and cementite in the pearlite eutectoid may be calculated as follows:

$$\% \text{ of carbon in pearlite} = 0.8$$

$$\therefore \ \% \text{ of } Fe_3C \text{ in pearlite} = 0.8 \times \frac{(3 \times 56 + 12)}{12}$$

$$= 0.8 \times \frac{180}{12} = 12$$

$$\therefore \ \% \text{ of ferrite in pearlite} = 100 - 12 = 88$$

Ratio of ferrite to cementite in pearlite $= 88:12 \approx 7:1$

The temperature at which the eutectoid reaction takes place is referred to as the A_1 point and is the same for all plain carbon steels. Below the eutectoid temperature the solubility of carbon in iron decreases still further until it is only about 0.007% at room temperature. As has been shown in figure 7.3, the addition of carbon to pure iron lowers the freezing point of the pure metal and also lowers the temperature at which f.c.c. iron changes to b.c.c. iron, i.e. lowers the A_3 temperature from 910°C for pure iron to 723°C for γ iron containing 0.8% carbon. Dilatometry, a technique mentioned previously, can again be used to determine the temperatures at which the crystal structure changes take place in a series of iron–carbon alloys of known carbon content. Thus the effect of carbon on the critical temperatures can be determined.

pearlite. Since this ferrite contains only a very small amount of carbon, the carbon content of the remaining austenite will increase progressively as more and more ferrite is formed, until at 723°C (the eutectoid temperature) the structure consists of ferrite, containing about 0.04% carbon, and austenite containing 0.8% carbon. At 723°C the remaining austenite transforms into the eutectoid (pearlite) by depositing alternate layers of ferrite and cementite. The temperature at which pearlite is formed is also known as the *lower critical temperature* and is the same for carbon steels of all compositions, i.e. NKL is a horizontal line.

The *upper critical temperature* of a steel varies with its carbon content and will be represented by a point on MKH; it is that temperature below which austenite begins to transform under slow cooling.

2. *Steel containing 0.8% carbon*

This alloy begins to solidify by depositing dendrites of austenite and, when solidification is complete, the structure will consist of cored austenite grains with average composition 0.8% carbon.

As the alloy cools slowly, diffusion causes the structure to become uniform and no further change takes place until point K is reached. For an alloy of this composition the upper and lower critical temperatures coincide, and the austenitic structure transforms at this temperature to become completely pearlitic.

3. *Steel containing 1.2% carbon*

This alloy will solidify in a similar way to the 0.8% carbon alloy by forming austenite grains with an overall carbon content of 1.2%. When the temperature falls to the upper critical temperature (point on line HK) pro-eutectoid cementite begins to precipitate at the austenite grain boundaries. Since cementite (Fe_3C) is being deposited, the remaining austenite will become less rich in carbon and, when the temperature falls to 723°C, the austenite will contain 0.8% carbon. Pearlite will now form giving a final structure of pro-eutectoid cementite and pearlite.

Thus, in an alloy which has been allowed to cool sufficiently slowly, so that diffusion has not been hindered, one of the following structures will exist:

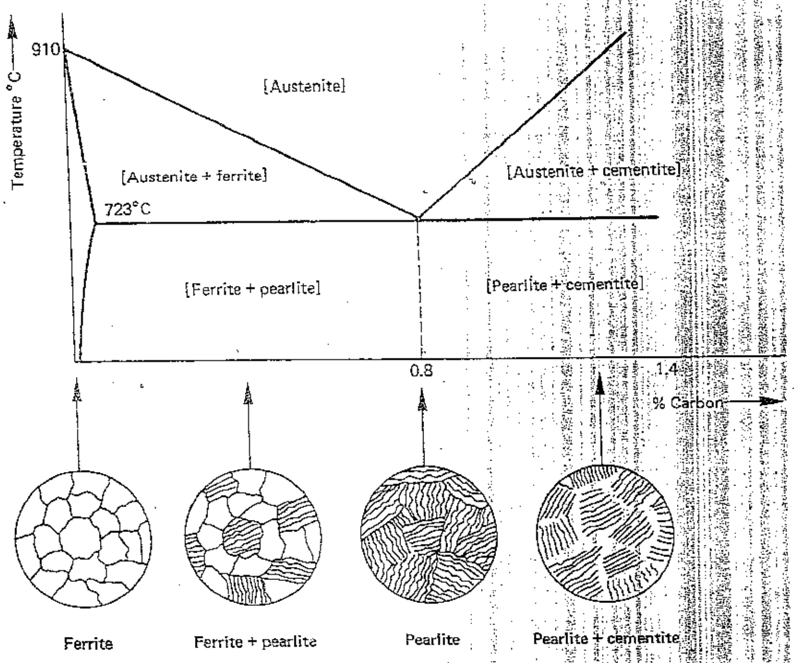
a) With less than 0.006% carbon it will be completely ferritic. Such an alloy would be classed as commercially pure iron or ingot iron. ("Armco" iron is such a material.)
b) Between 0.006% and 0.8% carbon the structure will contain ferrite and pearlite, the proportion of the latter increasing as the carbon content increases. In practice, with very low carbon contents (up to about 0.04%) some of the carbon may be present as films of free cementite at the ferrite grain boundaries.
c) At exactly 0.8% carbon the structure will be entirely pearlitic.

plain

4

7.4 P
Steel

Fig. 7.4
Micro-constituents of plain carbon steel'



d) Between 0.8% and 1.7% carbon the structure will consist of cementite and pearlite, in relative amounts which depend on the carbon content.

The above structures are illustrated in figure 7.4. The composition of the pearlite area in the microstructure of any plain carbon steel is always the same, namely 0.8% carbon. If the original carbon content of an alloy is either more or less than 0.8%, then this will result in a variation of the amount of either pro-eutectoid cementite or pro-eutectoid ferrite. Steels with less than 0.8% carbon are known as *hypo-eutectoid* steels, while those containing more carbon are referred to as *hyper-eutectoid* steels.

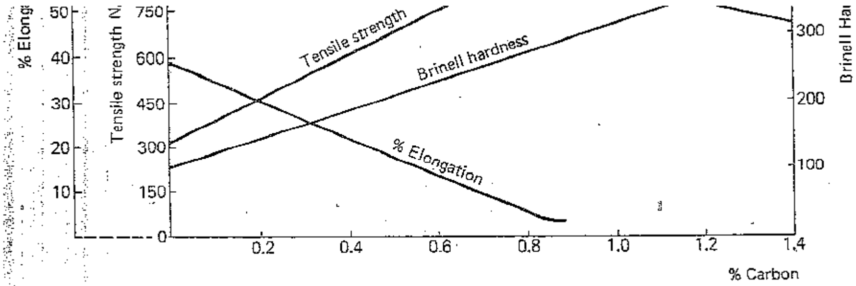
It follows from the above that microscopic examination of slowly cooled plain carbon steels can be used to estimate the approximate carbon content. For example, for a hypo-eutectoid steel

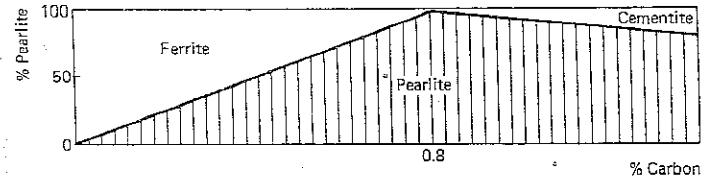
$$\% \text{ carbon} = \% \text{ pearlite} \times 0.008$$

7.4 Plain Carbon Steels

In simple terms, a plain carbon steel may be said to be an alloy of iron and carbon containing less than 1.7% carbon. In practice, however, these steels rarely contain more than about 1.4% carbon and other elements are also present, either as deliberate additions (e.g. manganese) or as impurities (e.g. sulphur and phosphorus).

a function of
composition and
structure





Mechanical Properties

Approximate values for the common mechanical properties of ferrite, pearlite and cementite are given below:

Micro-constituent	*Tensile strength*	*% Elongation*	*Hardness*
Ferrite	330 N/mm²	40	100 HV
Pearlite	900 N/mm²	5	270 HV
Cementite	—	—	650 HV

Hence the mechanical properties of slowly cooled plain carbon steels depend on the relative proportion of the micro-constituents present. The hardness, tensile strength and % elongation vary in an approximately linear manner with the carbon content up to 0.8%. Above 0.8% carbon, free cementite appears in the structure and the tensile strength starts to decrease (fig. 7.5). A simple empirical relationship between carbon content and tensile strength of hypo-eutectoid steels in the slowly cooled condition is given by

$$y = 700x + 350$$

where y = tensile strength in N/mm^2 and x = % carbon.

Killed and Rimming Steels

At the end of the refining stage of the steelmaking process, the liquid steel contains excess oxygen. If steel in this condition were cast, the oxygen would react with the carbon present to form carbon monoxide gas, which would cause the solidified steel to be porous throughout its whole mass. The oxygen may be completely removed by adding de-oxidising elements such as man-

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ganese, silicon or aluminium. The steel is then said to be **killed** because it is quiescent during casting.

A different type of steel ingot may be made by removing only some of the oxygen before casting. After partial deoxidation the liquid steel is run into ingot moulds and starts to solidify by forming a rim of pure iron adjacent to the mould walls. Then the remaining oxygen reacts with the carbon present to form monoxide gas.

$$FeO + C \rightarrow Fe + CO$$

Eventually the top of the ingot freezes over and the gas is then not able to escape so that blowholes are formed in the steel. These blowholes counteract the shrinkage which occurs when the steel solidifies. This type of steel is known as **rimming** steel.

7.5 Effects of Elements other than Carbon in Plain Carbon Steels

Due to the incomplete removal of impurities during steelmaking, and also because of deoxidation reactions, cast steel will contain both soluble impurities and insoluble non-metallic inclusions, all of which may be segregated, i.e. non-uniformly distributed. Hot working will cause these inclusions and segregated impurities to be "strung out" in the direction of working, thus producing a fibrous structure, giving a variation in property with direction, i.e. causing anisotropy.

In addition to carbon, plain carbon steels contain the following elements:

Manganese up to 1%	Sulphur up to 0.05%
Silicon up to 0.3%	Phosphorus up to 0.05%

The amount of an element reported in a chemical analysis can be misleading with regard to its effect on the properties of steel since the element may occur in a number of different forms. For example, manganese may be in solid solution and thus strengthen the ferrite, but may also be present as Mn_3C, MnS and MnO. The effect of each element present on the properties of the steel will therefore depend on the amount present and the form in which it occurs.

Manganese is an essential and beneficial element in nearly all types of steel. It is introduced as ferromanganese, the amount added varying with the type of steel required and with the degree of oxidation in the melt. The manganese acts as a deoxidiser by combining with dissolved oxygen. Sufficient must be added so that, after deoxidation, about 0.3 to 1% will remain in the steel. Manganese also reacts with sulphur, thus ensuring that sulphur is present as MnS and not as objectionable FeS. Manganese is soluble in austenite and ferrite and also forms a stable carbide Mn_3C. It raises the yield strength, tensile strength and impact resistance. It increases the depth of hardening following quenching, but also increases the tendency to crack during quenching, so that manganese should be below 0.5% in high carbon steel intended to be quenched.

since no ferro-silicon is used in the partial removal of oxygen. Silicon has little direct effect on the mechanical properties, but in high carbon steels it should not exceed 0.2% since it helps the breakdown of cementite into graphite and ferrite. Silicon also imparts fluidity to steels used for the production of castings.

Sulphur can exist in steel as MnS or FeS. FeS is soluble in liquid steel, but virtually insoluble in the solid state. There is, therefore, the possibility that FeS might precipitate as films at the austenite grain boundaries, and, indeed, this can occur with sulphur levels as low as 0.1%. Since FeS has a fairly low melting point, the steel would tend to crumble during hot working. Also, being brittle at ordinary temperatures, FeS would make steel unsuitable for cold working or even for service of any kind. To ensure that sulphur is not present as FeS, an excess of manganese must be present in the steel. This is achieved during steelmaking by adding enough ferromanganese and, provided that about five times the theoretical manganese requirement is added, MnS is formed in preference to FeS. The MnS is insoluble in the molten steel so that some enters the slag with the remainder being distributed as globules throughout the steel. The MnS has a high melting point and is plastic at hot-working temperatures, so that the globules become elongated into threads by hot rolling.

In general, the sulphur content of a steel should not exceed 0.05% and in high-quality material, including tool steels, should be below 0.03%. However, certain free-cutting steels contain up to about 0.3% sulphur with 1.5% manganese and a phosphorus content less than 0.03%. The MnS present then acts as a "chip-breaker" to bring about an improvement in machinability.

Phosphorus forms brittle iron phosphide Fe_3P, which is soluble in steel. In solid solution, phosphorus exerts a hardening effect, but it must be kept below 0.05% because of possible brittleness, especially if Fe_3P should appear as a separate constituent in the micro-structure. In hot-rolled steel any areas high in phosphorus due to segregation are often elongated into bands which are characterised by the absence of pearlite. Non-metallic inclusions (e.g. MnS) are often associated with these bands of ferrite. Such light-coloured streaks are sometimes known as "ghost-bands" and are a source of weakness because of the absence of pearlite.

7.6 Classification and Applications of Plain Carbon Steels

Plain carbon steels may be classified into three main groups:

1 Low carbon steel (mild steel) containing less than 0.3% carbon

2 Medium carbon steel containing 0.3 to 0.6% carbon

3 High carbon steel containing 0.6 to 1.4% carbon

Low Carbon Steels

Low carbon steels may be sub-divided into:

1 *Dead mild steel containing 0.05 to 0.10% carbon*
In the manufacture of this material, complete deoxidation is not carried out and it is made as rimming steel. It is produced as hot and cold worked sheet, strip, rod, wire and tube, and is available in the hot-worked or process annealed condition. The steel has excellent ductility, which is essential for the deep drawing of sheet, but the strength is low. This type of steel is used for the manufacture of motor-car bodies, refrigerators, washing machines, office furniture, pressings, rivets and nails.

2 *Carburising steels containing 0.1 to 0.2% carbon*
The strength and hardness of these steels is low, but a hard surface can be obtained by carburising (see Chapter 8) in order to combine a wear-resistant surface with tough core properties. The ductility makes rapid machining difficult unless the MnS content is increased by using a sulphur content of 0.2 to 0.3% with about 1% manganese.

3 *Constructional mild steels containing 0.2 to 0.3% carbon*
As the carbon content is raised above 0.2%, the strength increases into the range required for constructional purposes, but the ductility decreases. The fabrication qualities (working and welding characteristics) of this material are very good. The steel is hot-rolled into plate for making boilers, ships and vehicles, and also into sections (e.g. girders, beams, joists) for use in bridges and buildings. The steel is often used in the hot-rolled condition, but the smaller sized material may be normalised (see Chapter 8).

It has been estimated that about 90% (by weight) of all steel used is of this low carbon classification.

Medium Carbon Steels

Medium carbon steels are often quenched-hardened and fully tempered to give good strength with maximum toughness, which are the properties required of components such as shafts, gears and connecting rods. Steels containing 0.4 to 0.5% carbon may also be surface hardened by localised heating and quenching.

High Carbon Steels

High carbon steels are quench hardened and lightly tempered to give high hardness with only limited toughness. Material in this classification is mainly used for making fairly small, relatively inexpensive cutting tools so that the group as a whole is known as *carbon tool steel*. Steels containing 0.6 to 0.8% carbon are used for making dies, springs, wire ropes and railway tyres. Many different kinds of tools are made from higher carbon steels as indicated below.

1.0–1.2	Files, axes, saws, knives
1.2–1.4	Razors, drills, wood-cutting and metal-cutting tools

Limitations of Plain Carbon Steels

Plain carbon steels have many limitations including the following:

a) If reasonable toughness and ductility are required, the maximum tensile strength obtainable is about 700 N/mm^2.
b) Large sections cannot be effectively hardened, thus restricting their use to relatively thin sections.
c) Water quenching is necessary for full hardening with consequent risk of distortion and cracking.
d) Rapid softening above about 300°C limits their use for high-speed metal cutting.
e) Poor resistance to corrosion and to oxidation at elevated temperatures.

To overcome these limitations, additional elements are added to the steel to give alloy steels with specific properties. The main alloying elements include manganese, nickel, chromium, molybdenum, tungsten, vanadium, cobalt and silicon.

7.7 Cast Irons

When carbon separates from liquid or solid iron it may do so either as free carbon (*graphite*) or as cementite (Fe_3C), according to the conditions under which the deposition takes place and the nature and amounts of other elements present. A high carbon content, slow cooling and the presence of a significant amount of silicon all favour the formation of graphite. A low carbon content, rapid cooling and an appreciable manganese content help the formation of cementite. When more than about 0.4% silicon is present in an iron/carbon alloy, all the carbon present can be obtained in the form of graphite as a result of slow cooling. With less than 0.4% silicon this result cannot be achieved because of the difficulty of cooling at a sufficiently slow rate.

As stated previously, the iron/cementite phase diagram represents metastable equilibrium between iron and cementite; the stable system is iron/graphite. Just as the iron/cementite phase diagram represents the phase relations between iron and carbon when the carbon not in solid solution is present as cementite, so a diagram may be constructed (fig. 7.6) to represent the phase relations when the carbon not in solid solution occurs as graphite. The two diagrams, however, apply to very different conditions of cooling and composition. The diagrams are very similar, except that the solubility of graphite in iron is rather less than that of cementite, and also the eutectic and eutectoid points in the iron/graphite system are at a slightly higher temperature than the corresponding points in the iron/cementite system. In the

Fig.
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... general properties of a cast iron depend on the state of the carbon (whether free graphite or combined as cementite) and on the relative amounts, shape and distribution of these two states. The factors affecting the state of the carbon in cast iron include the following:

a) The rate of cooling from liquid to solid.
b) The presence of nuclei (e.g. *graphite*) in the liquid iron.
c) The chemical composition.
d) Subsequent heat treatment after the castings have been produced.

A **rapid cooling rate** tends to stabilise cementite, thus preventing the formation of graphite and giving a hard, white iron; a slow cooling rate helps the formation of graphite thus producing a soft grey iron. The cooling rate will depend on the section thickness and the type of mould material used, e.g. a metal mould allows much faster cooling than a sand mould. In castings of varying cross-section, the thinner section may consist of white iron, whereas thicker sections may be of much softer grey iron (fig. 7.7).

Fig. 7.7 Effect of section thickness on depth of chilling of grey cast iron

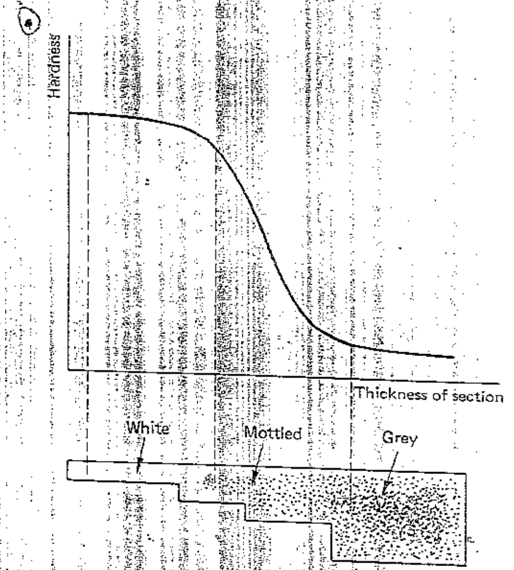
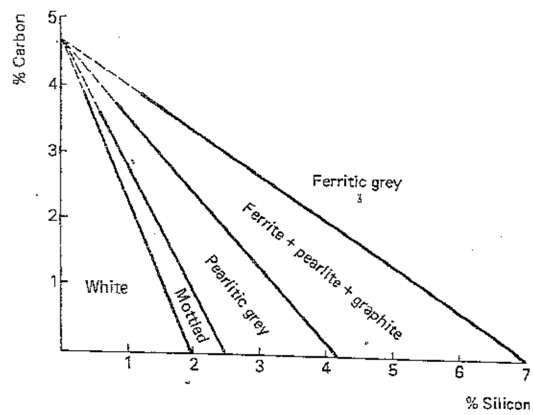


Fig. 7.8 Effect of carbon and silicon on the structure of sand cast irons



The **presence of nuclei** in the liquid iron exerts a big influence in promoting graphite formation. For example, the addition to the ladle of an *inoculant*, such as ferro-silicon, may cause an iron to solidify grey, whereas without the addition it would form a white iron. Conversely, the removal of nuclei (e.g. by superheating the liquid iron or keeping the iron molten for a long time) is a deterrent to graphite formation.

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Carbon lowers the melting point of the iron and increases the tendency for it to solidify as grey iron, thus favouring the production of a softer, weaker material. Silicon is a strong graphitising element and helps to ensure the production of grey iron. Sulphur has a considerable stabilising effect on cementite and other carbides. Manganese readily combines with sulphur and thus neutralises the carbide stabilising effect of the latter, and indirectly helps graphite formation. However, the direct effect of manganese is to stabilise the cementite, but this will only occur if the manganese content is greater than that required to combine with sulphur to form MnS. Phosphorus has little direct effect on the state of the carbon in cast iron. However, it increases the fluidity of the iron (by forming a low-melting-point eutectic) and lengthens the time taken to solidify. Therefore, if the prevailing conditions are graphitising, then phosphorus indirectly helps the graphitising by providing more time for graphitisation to occur.

The Carbon Equivalent Value

Figure 7.6 shows that a eutectic occurs in the iron/carbon system at 4.3% carbon. Therefore a carbon content greater than 4.3% would be expected to give a hyper-eutectic structure, whereas a carbon content less than 4.3% should give a hypo-eutectic structure. However, other elements, especially silicon and phosphorus, affect the composition of the eutectic point and a **carbon equivalent value** is used as an index to convert the amounts of these elements into equivalent amounts of carbon. A formula commonly used is

$$\text{Carbon equivalent value (C.E. value)} = \%\ \text{total carbon} + \tfrac{1}{3}(\%\text{Si} + \%\text{P})$$

In other words, the eutectic carbon content for a given iron is

$$4.3\% + \tfrac{1}{3}(\%\text{Si} + \%\text{P})$$

For example, an iron containing 3.5% carbon, 2.4% silicon and 0.6% phosphorus, might be expected to be a hypo-eutectic iron but when the effect of silicon and phosphorus are taken into account the iron becomes hyper-eutectic because its C.E. value is

$$3.5 + \tfrac{1}{3}(2.4 + 0.6) = 3.5 + 1.0 = 4.5\%$$

For a given cooling rate the C.E. value, therefore, determines how close a given composition of iron is to the eutectic point (C.E. value 4.3) and hence the amount of graphite likely to be present and the probable strength of a given section size. The C.E. value is also a useful guide to the tendency to form white iron ("chilling tendency") in thin sections (a low C.E. value promoting white iron formation) but pouring temperature and cooling rate also have a marked influence. The inter-relationship between carbon and silicon on the structure of sand-cast irons is shown in figure 7.8.

types of microstructures which a cast iron may have can be classified as follows.

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White Irons

Hypo-eutectic irons will solidify initially by the formation of austenite dendrites and the interdendritic areas eventually solidify as a eutectic of cementite and austenite (called ledeburite). Further cooling will cause the austenite to transform to pearlite (alternate lamellae of ferrite and cementite). At room temperature the structure is therefore one of primary dendrites of pearlite with interdendritic areas of transformed ledeburite (fig. 7.9).

Hyper-eutectic irons will solidify initially by the formation of needles of primary cementite so that at room temperature the structure will consist of massive primary cementite in a matrix of transformed ledeburite (fig. 7.10).

Grey Irons

In hypo-eutectic irons, austenite dendrites separate initially and will eventually be surrounded by eutectic cells in which austenite and graphite deposit simultaneously. The primary austenite and eutectic austenite become continuous and hence the structure consists of a dispersion of graphite in an austenite matrix.

In hyper-eutectic irons graphite (known as "kish" graphite) forms directly from the melt and is eventually surrounded by a eutectic of austenite and finer flakes of graphite.

Further cooling causes the austenite to transform to pearlite or to ferrite plus graphite or to a mixture of all three constituents. Ferrite is most likely to form with slow cooling rates, with high silicon levels or with high carbon equivalent values. The formation of a fully pearlitic matrix is favoured by fairly rapid cooling rates or low carbon equivalent values.

The microstructure of a grey cast iron with a mainly pearlitic matrix is shown in figure 7.11.

Mottled Irons

Such irons have intermediate structures between white and grey irons, so that the structure contains massive cementite and flakes of graphite with a pearlite matrix.

Phosphide Eutectic

Most cast irons contain phosphorus in amounts ranging from about 0.05% to 1.5%, so that another constituent, in addition to those already mentioned, may be present in the microstructure. Phosphorus exists as iron phosphide, Fe_3P, which forms a eutectic with ferrite in grey irons, and with ferrite and

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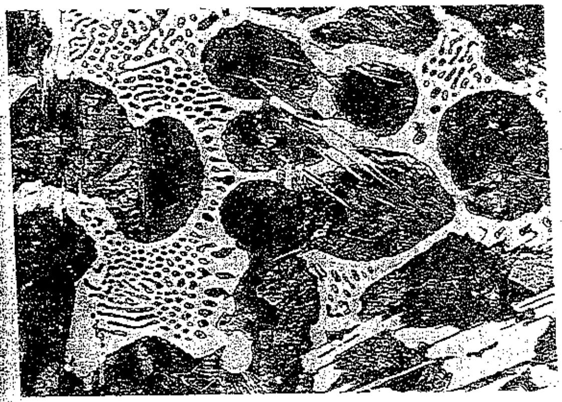

Fig. 7.9 Hypo-eutectic white cast iron; etched in 3% nital [× 300]

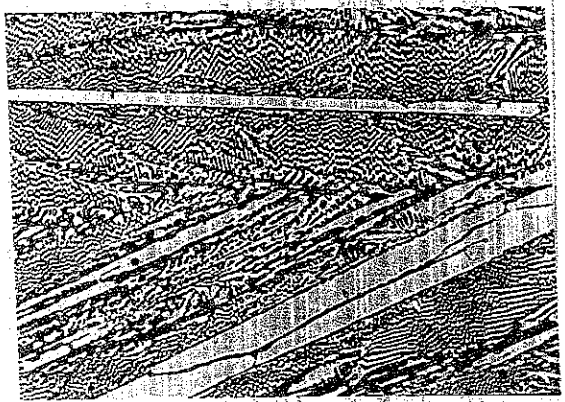

Fig. 7.10 Hyper-eutectic white cast iron; etched in 3% nital [× 300]

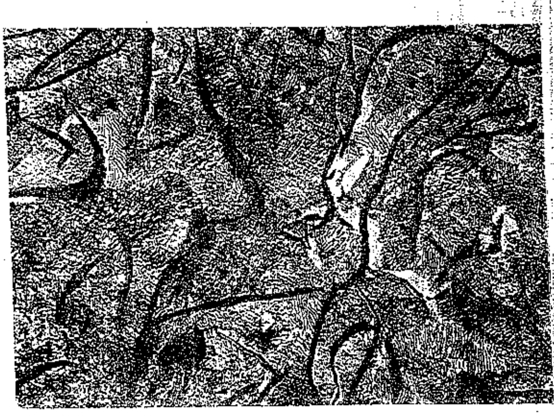

Fig. 7.11 Grey cast iron with medium-sized flakes of graphite and a mainly pearlitic matrix; etched in 3% nital [× 300]

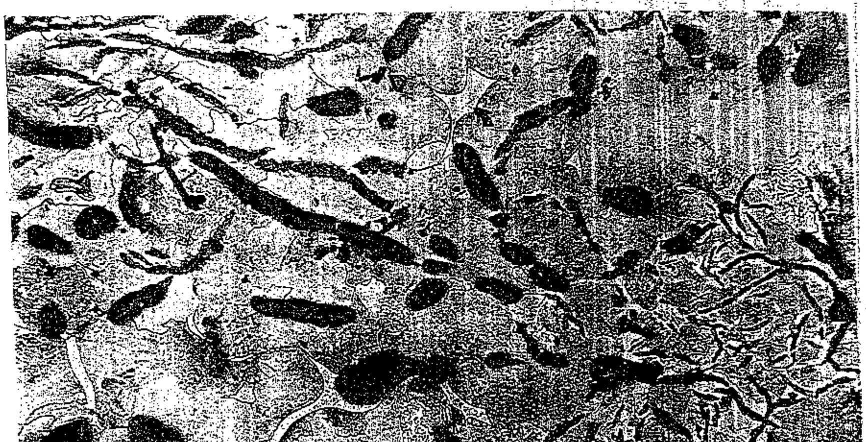

Fig. 7.12 Phosphoric grey cast iron showing islands of phosphide eutectic (arrowed); etched in 3% nital [× 300]

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... phosphide eutectic is also hard and brittle, but if it is present only in small amounts, it will not unduly weaken the iron. If, however, sufficient is present to form a network, it will have an embrittling effect. The phosphide eutectic is easily recognised in the micro-structure because of its "herring-bone" pattern (fig. 7.12).

7.9 Classification and Uses of Grey and White Cast Irons

Grey cast irons may be broadly classified into three main groups:

1 *Low-strength high-fluidity irons*
The high fluidity is achieved by using a high phosphorus content, up to about 1.5%, together with a high silicon content of 2.5 to 3.0%. The mechanical properties of the iron are not of great importance. The usage of this type of cast iron has dwindled considerably, although it is still used for such articles as manhole-covers.

2 *Engineering irons*
These irons must have reasonable strength with good castability. The type of matrix structure produced is controlled mainly by adjustment of the carbon and silicon contents in relation to the thickness of the casting and the type of mould used. The silicon content may be up to 2.5% for thin sectioned castings, but as low as 1.2% for thick sections. The sulphur and phosphorus contents are kept low because they tend to cause brittleness, but phosphorus cannot be decreased below about 0.3% because of the need for reasonable fluidity. The graphite flakes, though detrimental to the mechanical properties, ensure good machinability because they act as chip breakers, and their large bulk helps to improve castability by counteracting solidification shrinkage.

The British Standard Specification for grey cast iron is BS 1452, in which seven grades (10, 12, 14, 17, 20, 23 and 26) of material are specified with minimum tensile strengths of 155, 185, 215, 260, 310, 355 and 400 N/mm^2 respectively. Grades 10, 12 and 14 can be produced by casting the metal as tapped from the cupola.

3 *High duty irons*
For the higher tensile grades (i.e. grade 17 and above) the iron will require inoculation before casting or the addition of alloying elements. Inoculation may be carried out by adding a graphitising agent such as ferro-silicon or calcium silicide to the molten iron contained in a ladle. The effect of the inoculation (which disappears if the iron is re-melted) is twofold—it inhibits undercooling and consequent chilling or weak structure and also gives a better grain structure. The addition of elements such as nickel, chromium, molybdenum, and copper has some effect on graphite formation, but more particularly the strength of the matrix is increased.

The uses of grey cast irons are many and varied including the following:

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General engineering:	lathe beds and machine castings.
Automobile engineering:	cylinder blocks, brake drums.
Marine engineering:	pump cases.
Metal production:	ingot moulds, rolls.

White cast irons are very hard and unmachinable. Components are usually used in the as-cast condition after final grinding. A typical composition range would contain 2.5 to 3% carbon with 0.75 to 1.5% silicon. They are used in applications requiring an inexpensive material with high surface hardness, e.g. for earth-moving machinery, crushing and grinding equipment.

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7.10 Growth of Cast Iron

When a cast iron whose structure contains pearlite is subjected to prolonged heating above 700°C, the cementite breaks down into ferrite and carbon, thus resulting in an increase in volume. This expansion leads to cracking at the surface and allows hot gases to penetrate the minute cavities and oxidise the ferrite, causing a further increase in volume. The overall result is a stressing of the surface layers which causes a characteristic crazy cracking. This problem dictates that ordinary grey cast iron should not be used above 450°C. However, alloy cast irons which resist growth can be employed.

Stress Relief of Grey Cast Iron

Grey cast iron is usually stress-relieved by slowly heating to 500°C, holding at this temperature for a time dependent on casting thickness (1 hour per 25 mm), cooling to 300°C in the furnace, and then air cooling. It is best to limit the treatment temperature to 500°C in order to avoid grain growth, which is pronounced at about 550°C.

7.11 The Production of Malleable Cast Iron

Malleable cast iron is produced by the heat treatment (annealing) of white cast iron. The aim of the annealing treatment is to improve the ductility of the cast iron by causing the cementite (both the free Fe_3C and the eutectoid Fe_3C) to break down into ferrite and rosettes of temper carbon. The more rounded form of the rosettes avoids the stress-concentrating effects of flake graphite and also a softer matrix is produced, with little or no pearlite. The ductility of the iron, as measured by the % elongation of a tensile test piece, is increased from almost zero to between 5 and 15% as a result of the annealing.

Two types of material are made, known as Blackheart and Whiteheart malleable cast iron, but the annealing treatment in each case is rather a slow process. A further disadvantage is that, since a white cast iron has to be produced initially, the process is limited to small sections—usually less than about 5 cm thick. The British Standard Specification for Whiteheart iron is BS 309 (Grades 1 and 2) and for Blackheart iron is BS 310 (Grades 1, 2 and 3). These specifications lay down minimum values for yield strength, tensile strength and % elongation.

... composition of the initial white cast iron would be 2.5% C, 1.0% Si, 0.4% Mn, 0.08% S, 0.1% P. The former practice for the annealing treatment was to use batch-type furnaces and the total time taken was up to 4 or 5 days. After fettling (i.e. removal of unwanted metal by cutting and grinding), the white iron castings were placed in grey cast iron containers along with some non-reactive material (e.g. cinders) to act as a mechanical support. The lids were luted on with clay to exclude air and the containers loaded into the furnace. The castings were soaked at 900 to 950°C for a time dependent on the iron composition and the section thickness.

Modern practice is to use continuous furnaces in which a controlled non-oxidising atmosphere is circulated. The packing of the castings in inert material is dispensed with, and the whole annealing process is achieved in 48 hours or less. The continuous furnace may be of the moving hearth type in which the castings are slowly carried through a long zone of small cross-section.

The effect of the prolonged annealing is to cause the Fe_3C to break down, not into graphite flakes, but into rosettes of temper carbon. A fractured surface will thus be blackish in colour, hence the name "blackheart" * The final microstructure is entirely of ferrite with rosettes of temper carbon (fig. 7.13*a*) and is soft, readily machinable and nearly as ductile as cast steel. No oxidation of the carbon occurs during the annealing. Typical mechanical properties are

Tensile strength	350–400 N/mm^2
% elongation	10–20
Hardness	110–120 HB

Uses of blackheart malleable iron in the car industry include wheel hubs, brake shoes, rear-axle housings, pedals, levers and door hinges.

The annealing process need not be allowed to go to completion with the total decomposition of the pearlite. The process can be controlled to allow some pearlite to remain in the matrix, thus resulting in a harder, stronger material which is called *pearlitic malleable iron*. In this case the initial white iron is similar in composition to that used for making ordinary blackheart malleable iron, except that carbide–stabilising alloying elements are included to ensure the retention of some pearlite in the matrix. The final microstructure consists of rosettes of temper carbon in a mainly pearlitic matrix (fig. 7.13*b*). A tensile strength of 500 to 750 N/mm^2 with 2 to 10% elongation may be achieved and the product can compete with steel castings for some applications.

*Whiteheart Malleable Cast Iron

The initial white cast iron may typically contain 3.3% C, 0.6% Si, 0.5% Mn, 0.2% S, 0.1% P. Sulphur is a carbide-stabilising element and the presence of up to 0.2% makes the complete breakdown of Fe_3C very difficult. After fettling, the white iron castings are packed in a mixture of new and used

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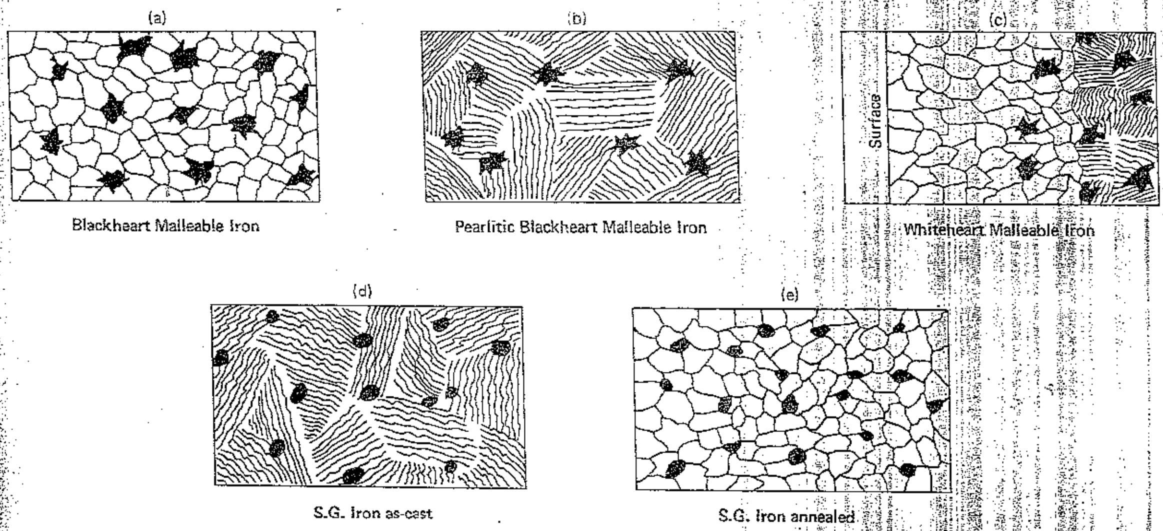

Fig. 7.13
Microstructures of various irons

haematite iron ore, in grey cast iron boxes which are sealed with lids. The boxes are then heated at about 1000°C for 3 to 4 days. As in the production of blackheart malleable iron, the Fe_3C present breaks down into temper carbon. In this case, however, the carbon formed at the surface of a casting is oxidised by contact with the iron ore and removed as CO_2. This causes more carbon to diffuse outwards from the core, and in turn this is removed by oxidation. Thus, after treatment a thin section may become completely ferritic, and on fracture gives a whitish appearance. Hence the name "whiteheart". Thicker sections will not be completely decarburised and so, while the outer zone is ferritic, this will merge into the interior which will contain some pearlite together with rosettes of temper carbon (fig. 7.13*c*). The amount of pearlite increases towards the centre of a casting. The surface of a casting may show some oxide penetration. Typical mechanical properties are

Tensile strength	350–450 N/mm^2
% elongation	5–10
Hardness	120–250 HB

Thin-sectioned components requiring good ductility may be made as whiteheart malleable castings, and uses include fittings for gas, water, air and steam pipes; bicycle and motorcycle-frame fittings; parts for agricultural and textile machinery.

Spheroidal Graphite Iron (S.G. Iron)

Cast iron with the graphite in nodular form is produced by inoculating the melt with a small amount of magnesium or cerium. Magnesium is usually used because it is cheaper and the addition is made as a nickel/magnesium alloy, resulting in about 0.05 to 0.10% Mg remaining in the iron. The composition of the molten iron is such that without inoculation it would solidify as white

Re-melting of the iron causes a reversion to flake graphite formation due to loss of magnesium.

The aim is to produce a final microstructure which consists of spheroids of graphite in a matrix of ferrite with only a little pearlite (up to a maximum of 10%). Such a structure is obtainable in the as-cast condition if the iron has a sufficiently low content of carbide and of carbide-stabilising elements. However, heat treatment may be necessary to achieve a completely satisfactory structure. The heat treatment is designed to ensure decomposition of carbide into spheroids of graphite. The micro-structures of as-cast and of heat treated S.G. iron are shown in figures 7.13*d* and *e* respectively.

Any heat treatment necessary must be carefully controlled in order to get the correct balance between temperature, time, section size and composition of the iron; the process usually involves two stages:

a) Austenitisation for a few hours at about 900°C in order to eliminate carbide.

b) Controlled cooling (e.g. about 50°C per hour) through the eutectoid temperature so that the change from austenite to ferrite and graphite spheroids takes place.

If the castings are free from carbide then a ferrite matrix may be achieved by sub-critical annealing only (i.e. ferritising). However, any carbide present will not be eliminated by such treatment and poor impact properties may result. During the heat treatment attention must be paid to the heating rate, the packing and support of the castings, and temperature distribution in the furnace, in order to avoid the production of warped or cracked castings. The advantages and disadvantages of heat treatment are summarised below.

Advantages

a) In addition to decomposing carbides into graphite spheroids, careful annealing will remove internal stresses and improve machinability.

b) Particular mechanical properties may be produced.

c) Castings of widely varying section may be made.

d) General foundry control need not be as precise as for producing S.G. iron for use in the as-cast condition.

Disadvantages

a) Escalating energy costs make heat treatment an expensive process.

b) Risk of distortion and/or cracking.

c) Increased cleaning costs unless scale-free castings are produced.

S.G. iron can be made in a range of compositions depending on the section size of the casting and the properties required. S.G. irons are not so section sensitive as grey irons, i.e. a large increase in section thickness only causes a small drop in tensile strength. Several grades of S.G. iron can be produced, the matrix of which may vary from a mainly pearlitic structure to one which is

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10 Non-ferrous Metals and Alloys

10.1 Copper and its Alloys

The wide use of copper and its alloys is due to its excellent electrical and thermal conductivity, good corrosion resistance, good working and alloying properties, and ease of soldering and brazing. It cannot, however, compete with steel or aluminium alloys where the strength/weight or strength/cost ratio is the critical factor, and it has been replaced by cheaper materials (e.g. aluminium alloys and plastics) in some applications.

Commercial Coppers

There are several grades of **commercial copper** available including both furnace-refined (fire-refined) and electrolytically-refined metal. The fire-refined material may be either de-oxidised or "tough-pitch" copper.

1 **Tough-pitch copper** contains 0.03 to 0.05% oxygen which is present as tiny globules of Cu_2O and which forms a eutectic with copper. The Cu_2O present has a negligible effect on most properties, but it may reduce the conductivity enough to make it unsuitable for some electrical work. The tough-pitch grade is the most satisfactory type of copper for working. However, when tough-pitch copper is heated to about 400°C in reducing atmospheres, it is prone to a defect called "gassing". The reducing gas (hydrogen or CO) penetrates into the metal and reacts with the Cu_2O present to form steam [$Cu_2O + H_2 \rightarrow 2Cu + H_2O$]. This steam cannot diffuse through the metal and it develops enough pressure to cause intercrystalline cracking of the metal. Reducing atmospheres may be involved in gas-welding operations and in tube making so that deoxidised copper is necessary for these applications.

2 **Deoxidised copper** is usually made by adding phosphorus (in the form of a copper/phosphorus alloy) to the copper before it is cast. The phosphorus causes deoxidation since it has a greater affinity for oxygen than has copper. Only a small excess of phosphorus (about 0.05%) remains in the copper after deoxidation, but this small amount is sufficient to reduce the electrical conductivity by as much as 25%. Therefore, this material is unsuitable for electrical purposes. It is also more difficult to work than tough-pitch copper.

3 **Oxygen-free high-conductivity** (OFHC) copper contains neither excess oxygen nor phosphorus and has high electrical conductivity. It is produced from electrolytically-refined copper which is re-melted and cast under controlled conditions and de-oxidised, if necessary, with lithium.

strength and much improved resistance to oxidation at elevated temperatures. Arsenical copper is used for making rivets and bolts and in chemical plant. It can be welded and is therefore suitable for fabrication into hot water tanks and steam plant.

5 **Free-machining copper**. The addition of 0.5% lead imparts free-machining properties to copper, but it decreases the electrical conductivity to less than 95% of that of pure copper. The addition of lead can also cause difficulty in hot working; lead is insoluble in copper, so that globules of lead are actually molten at hot working temperatures.

About 0.5% tellurium (or selenium) will also give free cutting characteristics to copper, while the electrical conductivity is significantly better than with the lead addition.

6 **Copper/cadmium alloy**. Copper for overhead wires is strengthened by alloying with 1% cadmium and then cold working.

Mechanical Properties and Uses

The **mechanical properties** of the various compositions of commercial grades of copper do not vary much, but the physical condition of the material has a marked effect. As cast the tensile strength is about 150 N/mm², which is increased by working and annealing to about 200 N/mm², with 50% elongation.

Strip and sheet may be sold in tempers. "Soft" temper refers to annealed material with a hardness less than 60 HV; "half-hard" material is slightly cold worked to 70 to 80 HV; and "hard" temper is more heavily cold worked to give a hardness of 100 HV or more.

Some **uses** of the various grades are given below:

a) In the electrical industry: wire for all kinds of conductors.
b) In the chemical and food industries: sheet and tube for construction of processing vessels.
c) In the building industry: sheet for roofing and making into water tanks; tube for water piping. END OF T1

10.2 Copper Alloys

The main elements which are added to produce copper alloys are zinc, tin, aluminium and nickel; other alloying elements used to a lesser extent include beryllium and chromium. Copper alloys are usually strengthened by solid solution hardening and work hardening, but some alloys can be precipitation hardened (e.g. copper/beryllium alloys). The more important copper-base alloys may be classified as follows:

Copper/zinc (Brasses)
Copper/tin (Tin bronzes)
Copper/aluminium (Aluminium bronzes)

Copper/nickel (Cupro-nickels)
Copper/beryllium (Beryllium-bronzes)

The binary systems relevant to the above alloys involve rather complex equilibrium diagrams, with appreciable initial solid solubility followed by intermediate phases. The initial copper-rich α solid solution has a f.c.c. structure and the alloys are soft, ductile and suitable for cold working. When the solid solubility limit has been exceeded, the β phase appears which has a b.c.c. structure and this increases the strength but reduces the ductility. Alloys containing the β phase cannot be easily cold worked but are suitable for hot working and are also used in the cast condition. Further addition of alloying element produces the γ and δ phases which are very hard and brittle.

10.3 The Brasses

Copper/zinc alloys are important engineering materials because of their ease of working, range of mechanical properties and good corrosion resistance. The part of the copper/zinc phase diagram which covers the commercial **brasses** is shown in figure 10.1, which also shows the variation of mechanical properties with composition. However, with normal cooling rates, the indicated solid state phase changes rarely reach completion.

Initially, zinc dissolves in copper to form the α-solid solution, thus strengthening the copper. The diagram shows that the solid solubility of zinc *increases* as the temperature falls to 453°C (which is unusual), then decreases slightly down to room temperature. The α solid solution has a f.c.c. structure and is soft and ductile, thus making it suitable for cold working. An interesting feature is that the ductility of the alloys increases with zinc content up to a maximum value at 30% zinc. The ductility then decreases with further addition of zinc until at 38% zinc (the limit of the α phase) it is similar to that of pure copper.

Alloys of composition between A and B solidify as α-solid solution whose grains are usually cored. Compositions between B and F initially solidify as α solid solution and then undergo a peritectic reaction, in which some of the α phase reacts with the liquid to form the β solid solution. On cooling, the amount of the β phase decreases as shown by the lines BG and FH. Between 453°C and 470°C, the β phase changes to a modified state referred to as β' due to the zinc atoms changing from a random to an ordered arrangement in the lattice. This change is shown by the dotted lines in the equilibrium diagram, but is of no practical importance, and the room temperature phase is usually referred to as β and not β'. Compositions between F and C form α phase which reacts at 905°C with liquid of composition C to form β phase. On cooling, α phase is precipitated from the β when the line FH is crossed and a Widmanstätten arrangement of α in β is produced. Compositions between C and E solidify as β phase. Alloys containing less than 46.5% Zn precipitate some α phase, while alloys containing more than 50% Zn precipitate γ phase on cooling.

Thus alloys containing 38 to 46.5% zinc have a micro-structure consisting of α and β phases, while those alloys having 46.5 to 50% Zn consist entirely of β

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alloy consists entirely of β. Thus alloys containing β cannot be easily cold worked and are used for castings and for hot working.

As already stated, brasses with more than 50% zinc contain the γ phase, which is a very hard and brittle constituent, so that the presence of the γ phase is avoided. The only commercial alloy which may contain the γ phase is the 50% zinc alloy which is used for brazing because of its low melting point.

Thus, to summarise, the addition of zinc to copper causes the following effects:

a) Increases the tensile strength (up to about 40% Zn).
b) Increases the ductility (up to 30% Zn).
c) Lowers the melting point.
d) Reduces the corrosion resistance.
e) Lowers the density.
f) Changes the colour from reddish brown to yellow.
g) Reduces the cost.

In the alloying of copper and zinc, the copper is melted first and then the zinc is added, allowance being made for the loss of zinc as vapour due to its low boiling point (907°C).

Classification of Brasses

The properties of a brass depend on the amounts of the phases present, so that commercial alloys may be broadly classified as:

1) α-brasses: very suitable for cold rolling into sheets, pressing, wire-drawing and tube-making.
2) α-β brasses: suitable for hot working by rolling or extrusion and for the manufacture of castings.

1 Alpha brasses

Annealed α-brasses can withstand heavy cold working without cracking. However, cold-worked material is prone to stress-corrosion cracking (called season cracking) unless stress relieving is carried out before use. If the material is to be hot worked then the amounts of low melting point impurities must be controlled, e.g. the bismuth and lead contents should be less than 0.002% and 0.02% respectively.

85-Cu/15-Zn alloy The colour of this alloy resembles that of gold and it is known as gilding metal. It has good corrosion resistance and is used for making imitation gold jewellery, compact boxes and also for plumbing fitments.

70-Cu/30-Zn alloy This alloy is notable for its ductility in cold working operations. It is known as "cartridge brass" because of its use in the manufacture of cartridge cases. It is used for cold-rolling into sheet, deep

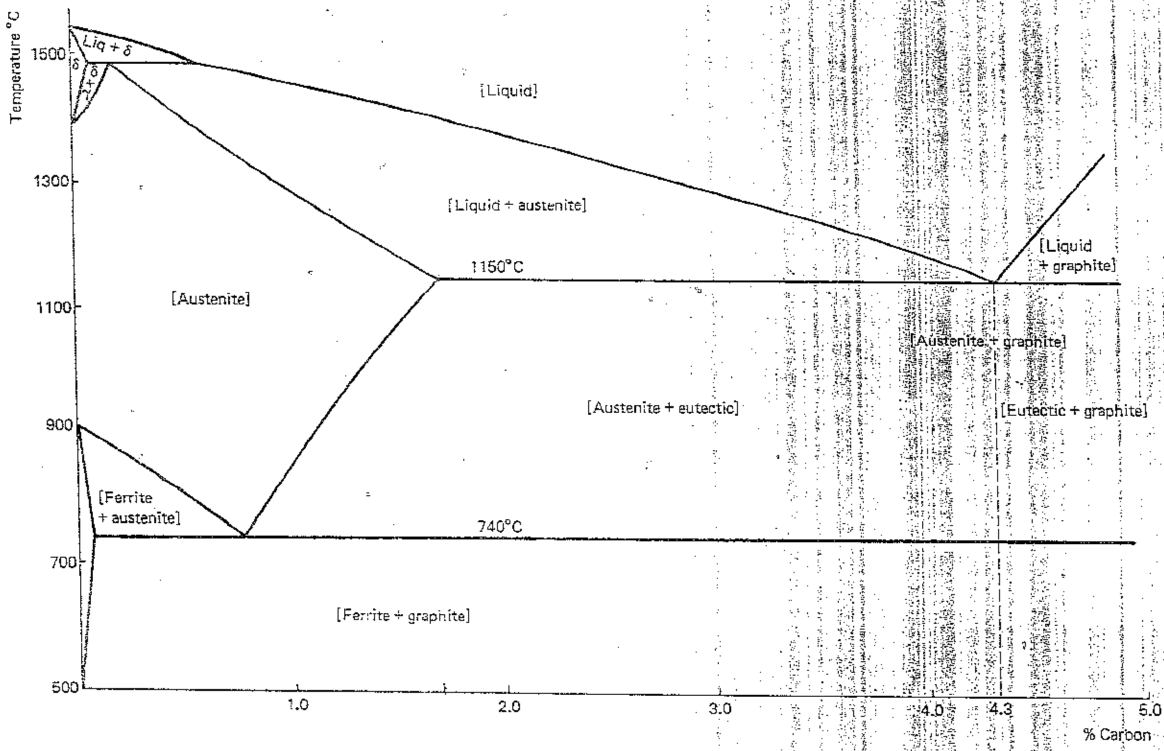

Fig. 7.6 Iron-end of iron/graphite equilibrium diagram

iron/graphite system it can be seen that the eutectoid reaction involves the transformation of austenite to ferrite and graphite, while the eutectic is composed originally of graphite and austenite; the latter subsequently changing into ferrite and graphite at the eutectoid temperature.

Cast irons contain about 2.5 to 4.5% carbon, so that the eutectic reaction, which occurs at 1150°C, plays an important part in the resulting microstructure. A cast iron may be classified as either hyper-eutectic or hypo-eutectic, depending on whether the carbon content is greater or less than 4.3% (the eutectic composition for plain irons).

A cast iron may solidify to produce a microstructure in which either

a) some or all of the carbon content is present as graphite; the resulting iron is known as **grey cast iron** because of the greyish fracture of a sand cast test bar, or
b) all the carbon content is present in the combined state as cementite, Fe_3C. The cementite may be free Fe_3C and/or mixed with ferrite forming the eutectoid, pearlite; the resulting iron is known as **white cast iron** because of the whitish appearance of the fracture of a sand-cast test bar.

Cast iron, like steel, always contains the five elements carbon, silicon, manganese, sulphur and phosphorus, but has a far wider range of composition; such a range might be carbon 2.5 to 4.5%, silicon 1.0 to 3.0%, manganese 0.4 to 1.0%, sulphur 0.05 to 0.2%, phosphorus 0.05 to 1.5%.

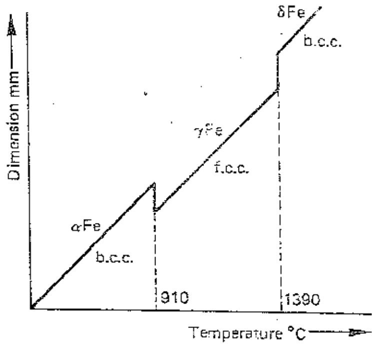

Fig. 7.2 Change in dimensions of pure iron with temperature

The atoms in the f.c.c. lattice are more densely packed together than in the b.c.c. form, so that in the change (on heating) from α iron to γ iron there is a contraction in volume; similarly when the reversion to the b.c.c. form takes place at the γ to δ critical point there is an expansion (fig. 7.2).

The critical points are referred to as follows:

the $\alpha \rightarrow \gamma$ change temperature is termed the A_3 point

the $\gamma \rightarrow \delta$ change temperature is termed the A_4 point.

Experimental determination shows that the temperatures at these change points are significantly different when heating cycles are used than when cooling cycles are employed. Thus, it is usual to refer to the change points obtained on heating as A_c points ("c" from chauffage), and those on cooling as A_r points ("r" from refroidissement). For example, the A_{c_3} and A_{r_3} points may be 935°C and 910°C respectively.

Another change which takes place in iron during heating is the loss of magnetism at 769°C (known as the Curie point). This temperature is referred to as the A_2 point, but no crystal change is involved and it is of no importance in heat-treatment.

The dimensional changes which take place during the alterations in crystal structure may be used as a basis for determining the temperatures at which the structural changes occur (i.e. the **critical temperatures**). The length of a small specimen of the iron is accurately measured over a known range of temperature using an instrument called a *dilatometer*. A graph of length as a function of temperature is plotted and the critical points are clearly apparent (fig. 7.2). Very slow heating and cooling rates must be used in order to minimise errors due to thermal lag.

7.2 The Iron/Carbon Phase Diagram

The addition of carbon to iron lowers the freezing point of the metal and also alters the temperatures at which the changes in crystal structure occur. Body-centred cubic iron has only very slight solid solubility for carbon to form a solid solution ferrite at low temperatures and δ-ferrite at high temperatures. Face-centred cubic iron can dissolve up to 1.7% carbon to form a solid